



University of Notre Dame

Department of Computer Science and Engineering

CSE 60745

Graph Learning: Techn & Apps

Assignment Date:	
Submission Date:	
Student Name:	



* This assignment contains 70 points. You are encouraged to finish independently. Any submitted work that is copied from any source or too similar to be an independent write-up will not be given credit. Please submit this homework (as a PDF or Word file including the cover page) to ND Canvas by **11:59pm on Wednesday, 9/30/2026**.

In this class, we have two-fold course objectives. The first one is to explore recent core techniques and advances in graph learning research, in the forms of class lectures, selected paper presentations and discussion, covering the topics of i) graph embedding techniques, ii) graph neural networks, iii) knowledge graphs, and iv) advanced topics on graph learning. Through our learning, we want to summarize the benchmarks and baselines that have been widely used for graph representation learning for 1) homogeneous graphs, 2) heterogeneous graphs; and 3) knowledge graphs.

Problem 1 (35 pts). Based on the above context, you are asked to summarize the *benchmarks*, *baselines*, and commonly used *evaluation metrics* for graph embedding methods with homogeneous graph structures for the task of node classification and link prediction as comprehensive as you can. *Note:* (1) For the benchmarks, please describe each of the benchmark graphs (e.g., [Cora](#)) and draw the related graph schema. (2) For the commonly used baselines, please provide the reference (i.e., original paper) of each baseline and briefly describe the method.



Problem 2 (35 pts). Based on the above context, you are asked to summarize the *benchmarks*, *baselines*, and commonly used *evaluation metrics* for graph embedding methods with heterogeneous graph structures for the task of node classification and link prediction as comprehensive as you can.

Note: (1) For the benchmarks, please describe each of the benchmark graphs (e.g., [DBLP/ACM](#)) and draw the related graph schema. (2) For the commonly used baselines, please provide the reference (i.e., original paper) of each baseline and briefly describe the method.